FAIR Terminology Meets CLEAR Global

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Abstract. This paper presents the design and implementation of a web application aimed at facilitating access to a multilingual terminology database specifically created for speakers of marginalized languages that often cannot access information they need and want in a language and format they understand. The database contains essential concepts translated into 39 languages, for a total of 9,506 terms. The design of the database and the web application has been performed in order to follow the FAIR principle of Open Science in order to guarantee efficient data organization and retrieval. The web application, developed using the R Shiny framework, enables users to search the database by selecting a source language and, optionally, a target language. The interface displays concept definitions in English and provides filtered lists of equivalents in the chosen languages. Users can export selected datasets in TermBase eXchange standard (TBX) format for reusability. The interface offers a user-friendly experience, allowing users to navigate through the database effortlessly.

Keywords: language technology in crisis situations · humanitarian termbases · FAIR terminology

1 Introduction

The ability to access vital information across languages is fundamental for effective communication and understanding in an interconnected world and even more in situations where the possible miscommunication in critical situations can worsen the severity of the event and retard effective response efforts. For example, during a crisis, there is often a flood of information from various sources, including official channels, social media, and person-to-person. Sorting through this abundance of information can be overwhelming for both responders and affected individuals, leading to confusion and conflicting messages. Crisis response efforts involve multiple organizations, agencies, and stakeholders operating in dynamic

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and high-pressure environments. In multicultural societies or international crises, language and cultural differences can delay and interfere effective communication between communities. Misinterpretation of messages, inability to understand instructions, and cultural sensitivities may hinder the dissemination of critical information and the provision of assistance. Finally, infrastructure damage or overload of communication networks during crises can disrupt traditional communication channels, such as phone lines or internet access. This can limit the ability of responders to disseminate information and for affected individuals to seek help or access emergency services. Addressing these challenges requires proactive measures, including robust communication strategies, interoperable communication systems, and linguistically sensitive messaging.

Terminology databases play a crucial role in this context as a step to overcome linguistic barriers by providing comprehensive collections of essential concepts and their designations in different languages. This paper presents the design and implementation of a web application aimed at facilitating access to a multilingual terminology database, catering to diverse linguistic needs. The term base was originally developed by CLEAR Global in collaboration with its partners, and over the years it was translated into more and more languages[1].

The foundation of our research lies in the creation of a structured database adhering to FAIR (Findable, Accessible, Interoperable, Reusable) principles [14] and ISO standards, ensuring efficient organization and retrieval of multilingual terminological data.

1.1 Background

Translators without Borders (TWB/CLEAR Global) is a prominent organization in the translation sector, positioning itself as a charity while operating with a corporate structure. TWB is a part of CLEAR Global, a US-based nonprofit and operates on an asset-centered, platform-based model, partnering primarily with major industry players, but also with small local NGO's. Its mission is to help people get vital information and be heard, whatever language they speak. The objectives of TWB include facilitating volunteer translation services on a large scale and leveraging technology to streamline translation efforts [2]. By mobilizing over 100,000 volunteers and utilizing a platform-based model, the organization provides language services and solutions, helping partner organizations improve communication with marginalized communities, particularly in crisis situations and humanitarian contexts[2].

The discussion presented in [4] emphasizes the vital role of language in the well-being and safety of refugees, specifically Ukrainian refugees. It stresses the importance of tailored communication to meet language preferences and sensitivities, necessitating data on language preferences. It also recommends basic staff training and collaboration with language service providers to enhance communication and information access for refugees, underlining the significance of

[1] https://clearglobal.org/safeguarding-and-psea/.

[2] https://clearglobal.org/mt-rwanda-pushing-forward-language-tech-in-kinyarwanda/.

language support in crisis situations for ensuring effective support delivery and refugee integration.

A similar problem has been presented by [5] where the authors propose a Participatory Design approach used to investigate technology's potential benefits for refugees in Malaysia. This work emphasizes the need for mobile applications to prioritize aspects like privacy, safety, social connections, skill development, and language translation for their digital empowerment and facilitate their integration and well-being in new environments.

However, as shown in [13], it is estimated that over 90% of the world's languages lack adequate language technology support, affecting billions of people. These 'low-resource languages' in computer science, suffer from a scarcity of training text required for machine translation models. This imbalance in language technology amplifies the dominance of already well-supported languages.

In [1], the authors discuss existing models for student translation volunteering due to ethical concerns with for-profit initiatives and limitations of non-profit organizations, which restrict participation to trained translators. The authors propose Action Translate, a platform enabling student volunteering by utilizing machine translation for a streamlined experience. Recommendations include a volunteer-centric approach focusing on the agency, accountability, and collaboration to create sustainable platforms leveraging students' multilingual skills and motivations.

The question of how to realize these language resources in a sustainable way is proposed by [7]. In this work, the author proposes to develop machine translation (MT) systems for low-resource languages and he emphasizes the ecological and social impacts of mainstream MT systems developed by large corporations, advocating for alternatives, independently operated "low-tech" MT solutions. Aligning with indigenous values, sustainable MT workflows prioritize inclusivity, collaboration, and environmental stewardship, promoting the preservation and promotion of indigenous languages and knowledge.

Finally, in [8], the author discusses how in the context of major emergencies like eruptions, earthquakes, and epidemics intense debates have been sparked between the government and society, significantly impacting urban planning and daily life. Specifically, the project Datus-Gost aims to compile these impacts into a database, providing a new resource for the scientific community to study the history of pandemics and natural disasters in Sicily.

2 Description of Original Dataset

CLEAR Global developed the PSEA (protection from sexual exploitation and abuse) and safeguarding glossary with international organizations such as the International Organization for Migration, the Safeguarding Resource and Support Hub, the CDAC network, CHS Alliance, UNICEF, and other members of the Inter-Agency Standing Committee. The 208 concepts and their English definitions were selected and agreed upon by the representatives of these organizations in 2021. The first target languages to be included in the glossary were also requested and funded by these organizations. Subsequent expansion into new target languages has been necessitated by crises happening around the world. The floods in Pakistan prompted 5 new languages, the war in Ukraine brought about the addition of 10 Eastern European languages, the earthquake in Turkey made Turkish necessary (as Arabic and Kurdish languages were already included)[3].

The glossary is intended to be used predominantly by humanitarian organizations and their workers, employees or volunteers. The need to expand the PSEA glossary to new languages for disaster response is explained by the fact that sexual exploitation and abuse become an issue in any crisis context in which humanitarians respond. Once a community has been affected by a crisis, it increases their vulnerability, changes power balances, and disrupts normal forms of protection. So the risk that humanitarians arriving to support in natural disasters will abuse citizens is no different from the risk in a conflict situation. PSEA is a sensitive topic, which is made even more complicated by the fact that many aspects of gender identities and sexual minorities are not discussed in certain languages. This means that there is no equivalent term in the target language, therefore the creation of a new term candidate or a paraphrase was necessary. The terminology work was done with the help of TWB's community members who speak the given languages. Each term candidate was reviewed by another community member, and discussions of problematic terms started at this step. Once an agreement was reached, the list of terms and definitions was reviewed by a subject-matter expert. The most challenging terms were also field tested, i.e. a researcher conducted focus group discussions checking the understanding of selected terms with local speakers of the language[4]. The process often involved several iterations of the steps to reach a consensus and took several months, sometimes over half a year for hard to source languages or languages without a standardized orthography. The collaborative work was carried out in Google spreadsheets, which allows the addition of as many columns for equivalents as necessary, and the final data were stored in TermWeb 4[5].

The original raw data consists of a table of 208 concepts (rows) and 39 languages. A small extract of the file is shown in Fig. 1. For each concept there is a unique identifier assigned by TermWeb 4 and a definition of that concept in English (columns 'Concept ID' and 'PL Definition' respectively). For each language, there are six columns that describe: the term that designates the concept in that language (the name of the first column refers to the language), the identifier of that term, the region (such as United States or United Kingdom), the type (such as full form or acronym), the definition of the concept translated in that language, and whether the term is the head term for that concept (in case there are multiple equivalents for that concept and language). For a concept, there may be alternative equivalents for a given language. This sums up to a total of 683 columns.

[3] For a detailed history, see https://clearglobal.org/safeguarding-and-psea/.

[4] https://translatorswithoutborders.org/resource/pseah-terminology-iraq-report.

[5] https://datcatinfo.termweb.eu/search/terms.

	Concept ID	PL Definition	Amharic	Term ID	Region	Term type	Definition	Head term	Arabic	Term ID
2	1778	A deliberate act with actual or potential negative effects upon a person's safety, well-being, dignity, and development. It is an act that takes place in the context of a relationship of responsibility, trust or power.	ጥቃት	1778-8			በሰው ደህንነት፣ ጥሩ ኑሮ፣ ክብር እና እድገት ላይ አሉታዊ ተጽእኖዎች ያስከተለ ወይም ሊያስከትል የሚችል ሆን ተብሎ የተደረገ ድርጊት። ከኃላፊነት፣ ከእምነት ወይም ከስልጣን ግንኙነት ዓውድ አንፃር የሚፈጸም ድርጊት ነው።	True	اعتداء	1778-9
3	1779	An obligation or willingness to use power responsibly and be held accountable for one's actions, both as individuals and as organizations.	ተጠያቂነት	1779-8			እንደ ግለሰብም ሆነ እንደ ድርጅት ሥልጣንን በኃላፊነት ለመጠቀም እና ለማንኛውም ድርጊት ተጠያቂ የመሆን ግዴታ ወይም ፈቃደኛነት።	True	مساءلة	1779-2
4	1781	A proposed strategy or course of action.	የድርጊት መርሐ-ግብር	1781-8			በሃሳብ ደረጃ የቀረበ ስልት ወይም ተግባራዊአካሄድ።	True	خطة عمل	1781-9

Fig. 1. A small extract of the original tabular data. The entire table consists of 208 rows and 683 columns for a total of 9,506 terms in 39 languages.

Fig. 2. A screenshot of the original Web application running at CLEAR Global

As an example: the concept identified with the sequence '1781' is defined as 'A proposed strategy or course of action'. The designation of this concept in Amharic language is identified with the sequence '1781-8' which is also the head term for this language. The Arabic equivalent is identified with the sequence '1781-9'. A total of 9,506 terms are documented in this database.

A screenshot of the Web application that is currently running on CLEAR Global servers[6] is shown in Fig. 2. The visualization of the data follows the structure of the original table with additional filtering functionalities. The equivalent of the English term in the other languages are shown as a list of vertical columns that can be scrolled horizontally. Additional information on the term (definition, source, usage note, etc.) can be seen by clicking on the term. One important element that is added to this application is the possibility to listen to the pronunciation of the term for some languages (for example, in Fig. 2 Amharic, Arabic, Bengali, and Czech have this feature active).

[6] https://glossaries.clearglobal.org/psea/

3 Designing the Database

In order to reshape the dataset and design a database that follows the FAIR terminology paradigm [9,10,12], we re-engineered the abstract structure of a possible database that represents these data according to the ISO 16642: 2017 [6]. This standard specifies a concept oriented three-tier hierarchical structure for a terminology database, where the three levels are: Concept Entry which is "information that pertains to a single concept", Language Section which is a "container for all the term sections of a concept entry for a given language, as well as information pertaining to the concept in that language", and Term Section which "contains exactly one term, and information about the term".

We use the entity-relationship (ER) model [3] to design the abstract representation of the terminology database that, in this project, will be subsequently transformed into a relational database. The basic elements of this schema include: entities, which represent the set of objects of interest that have common properties; properties, which describe the attributes or characteristics of these entities; relationships, which delineate a logical link among entities with a corresponding pair of numbers (the cardinality (n, m)) that indicate the minimum and maximum number of elements that can be involved in this relationship.

The Entity-Relationship (ER) schema in Fig. 3 can be read as follows:

- The entity "Concept Entry" is identified by its own identifier and can contain multiple language sections (cardinality (0, n)). The definition of the concept can be interpreted as the formal definition using a formal language (not a natural language);
- The entity "Language Section" is identified by the pair "Concept" through the relationship 'contain' and one identifier - in our case it will be the ISO language code. A language section is associated with the "Term Section" entity and can include different terms (cardinality (0, n)) which will be considered synonyms as they designate the same concept. The definition at the Language Section level can be interpreted as the natural language definition of that concept in that specific language.
- The entity "Term Section" is identified by its own identifier and can be included in only one "Language Section" (cardinality (1, 1)).

4 Implementing a Web Application

The design of the Web Application has followed two main principles: i) to allow users to search the multilingual terminology database with a source language and an optional target language; ii) to allow reusability of the database by means of standard serialization of the data into the TermBase eXchange standard (TBX). In particular, the application has been designed and implemented extending the R Shiny framework for web interactivity proposed by [11]. Compared to the

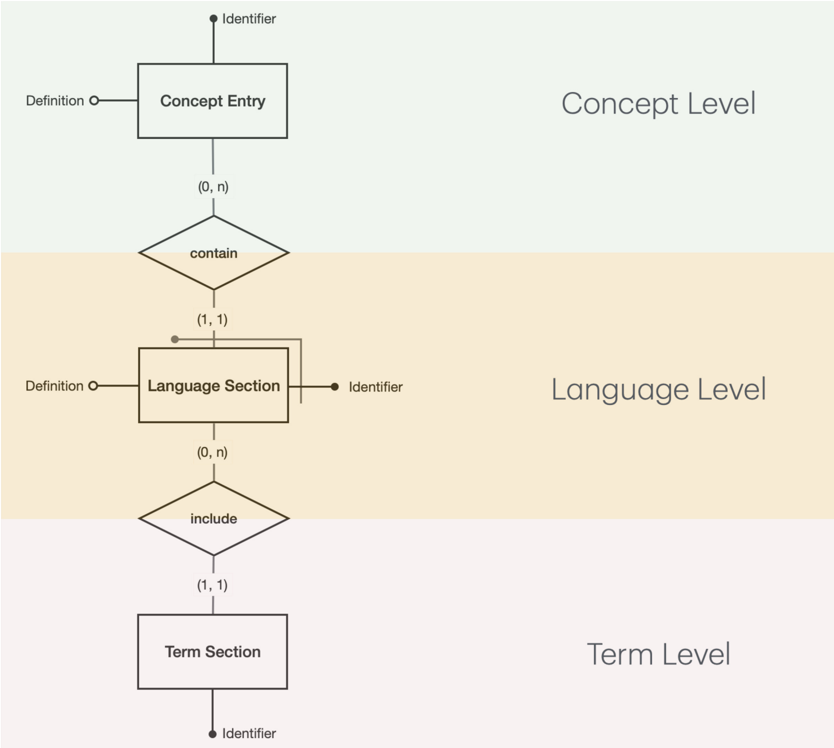

Fig. 3. The ER schema of the terminology database.

previous solution, our application includes the datatable (DT) package[7] to review the annotation, and the textplot package[8] for text visualization and analysis.

The current interface of the Web application[9] is shown in Fig. 4. A sequence of steps to interact with the interface is the following:

1. The user selects the source language and the term to analyze.
2. The interface displays the English definition of the concept designated by the chosen term.
3. The interface displays all the equivalents for that term in the chosen source language. The user can perform an additional filter of the displayed rows by using the "search" box.
4. The user can choose a target language for the chosen term.
5. The interface displays all the equivalents for that term in the chosen target language. The user can perform an additional filter of the displayed rows by using the "search" box.
6. The user can export the selected dataset as a TBX file.

[7] https://cran.r-project.org/web/packages/data.table/
[8] https://cran.r-project.org/web/packages/textplot/
[9] https://shiny.dei.unipd.it/CLEARGlobal_FAIR_psea

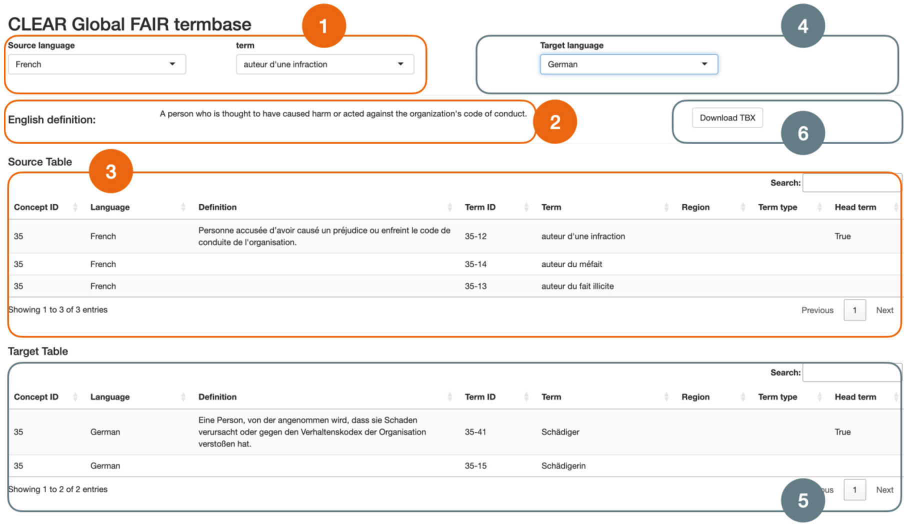

Fig. 4. Screenshot of the current Web application to browse and download the terminology database.

Besides the organization of the visualization from a different perspective compared to the original application, an optional bilingual terminological view compared to a fully multilingual parallel comparison, we want to highlight once more the importance of the organization of the data using international standards, like TBX. This feature will allow the possibility to broaden the interoperability and the reusability of the data produced as well as the compatibility with various language technologies and terminological tools, facilitating integration with other systems and organizations, such as the IATE terminology management system[10].

Finally, the open source code of the application as well as the full data (that will be in any case downloadable by the application) is available online[11].

5 Final Remarks and Future Works

This research presents a step towards facilitating access to vital information across languages through the development of a multilingual terminology database that supports the FAIR principles, aligning closely with the objectives of EOSC[12] initiatives like FAIRimpact[13], and is compliant with ISO standards as well as a user-friendly web application Through the implementation of the application, users can easily search, browse, and export terminological data, thereby promoting knowledge dissemination and linguistic inclusivity.

10 https://iate.europa.eu/.
11 https://github.com/gmdn/TPDL2024/.
12 https://eosc.eu
13 https://www.fair-impact.eu

This multilingual termbase is a global resource to equip people working to protect those at risk of sexual exploitation, abuse or harm to make PSEA (protection against sexual exploitation and abuse) communication more accurate, consistent and appropriate to their context. With the right words to talk about sensitive or traumatic issues, aid organizations can better ensure that people, especially marginalized people, are being listened to and receive the support they need.

In this context, we are currently planning the future steps in the development of this research that will focus on enhancing the database by improving the input and validation processes for new terms, incorporating user feedback to iteratively improve the interface, and experimenting with adaptable interfaces for mobile devices. This includes developing user-friendly interfaces for contributors to submit new terms and definitions, as well as implementing robust validation mechanisms to ensure accuracy and consistency of the added data. Integration of machine learning techniques could aid in automating the validation process, improving efficiency and scalability. Conducting a comprehensive user study will be crucial to gather feedback and insights for enhancing the usability and effectiveness of the web application interface. We will also plan to evaluate the usability and accessibility of the web application in challenging situations, such as limited screen size and varying network conditions. This would also include scenarios where internet connectivity may be unreliable or unavailable which means developing mechanisms to store essential data locally on users' devices, allowing them to access and interact with the terminology database even when offline. These efforts aim to further enhance accessibility, usability, and inclusivity, ultimately contributing to the goal of facilitating cross-cultural communication and knowledge sharing.

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